MIL-P-63293B (AR)
AMENDMENT 5
10 January 1986

SUPERSEDING
AMENDMENT 4
14 June 1985

MILITARY SPECIFICATION

PROJECTILE, 105MM, HEAT-T-MP, M456A2
METAL PARTS

This amendment forms a part of Military Specification MIL-P-63293B (AR), dated 18 October 1983, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1: Delete "MIL-C-60536 - Coating, Anodic, Hard, For Aluminum and Aluminum Alloys" and substitute "MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys"

PAGE 5

4.3.1: Delete "Plug and disc assembly....8597489.....20".

PAGE 6

TABLE I: Category I, delete "9. 8597489-Plug and Disc Assy.....20.....3.1.....4.4.2.9"

PAGE 8

* 4.3.4, 4.3.4.1, 4.3.4.2, 4.3.4.3 and 4.3.4.4: Delete these paragraphs in their entirety and substitute,

"4.3.4 Abrasion resistance test. Two fin and boom assemblies (or two fins and two booms) shall be subjected to the abrasion test in accordance with MIL-A-8625. If the actual parts can not be used during this test, then four test panels each, made in accordance with MIL-A-8625 and having the same alloy composition and heat treatment as the fin and boom shall be processed concurrently during the hard

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coating process and later be used for the abrasion test. If any component or test panel failed to meet the abrasion requirement in accordance with MIL-A-8625, the first article will be rejected."

PAGE 11

4.4.2.1:

Major Defect 101, after "Gage" add "(Total thread length must be inspected)"

Major Defect 103, after "Gage" add "(Total thread length must be inspected)"

* Major Defect 124, Delete "4.4.3.3.....3.3.....4.5.4.2" and substitute "4.4.3.3.....3.3.....4.4.3.3"

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4.4.2.2: Major Defect 101, after "Gage" add "(Total thread length must be inspected)"

PAGE 15

* 4.4.2.2: Major Defect 119, delete "4.4.3.3.....3.3....4.5.4.2" and substitute "4.4.3.3.....3.3.....4.4.3.3"

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* 4.4.2.4:

Major Defect 111, after "etching bath" add "(if applicable)."

Major Defect 112, after "mechanical abrasion" add "(if applicable"

PAGE 20

* 4.4.2.6: Major Defect 111, delete "4.4.3.3....3.3.....4.5.3.2" and substitute "4.4.3.....3.3.....4.4.3.3"

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4.4.2.7: Major Defect 120, after "Gage" add "(Total thread length must be inspected)"

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4.4.2.9: Delete this paragraph in its entirety.

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* 4.4.3.3 and 4.4.3.3.1 and 4.4.3.3.2: Delete these paragraphs in their entirety and substitute,

"4.4.3.3 Fin and boom abrasion resistance test. One fin and boom assembly (or set of components thereof) from each eight hours production or fraction thereof from the protective coating process shall be subjected to this test to determine compliance with MIL-A-8625. If the actual parts can not be used during this test, then two test panels each, made in accordance with MIL-A-8625 and having the same alloy composition and heat treatment as the fin and boom, shall be processed concurrently during the hard coating process and later be used for the abrasion test. Failure of any assembly, component, or test panel to comply with the abrasion requirement in accordance with MIL-A-8625 shall be cause for rejection of the product processed since the last acceptable test was performed."

PAGE 37

* 4.4.3.6: Delete "Adhesion testing" and substitute "Adhesion testing (if applicable)." At the end on the paragraph, add "The torque test can be voided if the seals are to be mechanically abraded after the application of the etching bath."

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* 4.4.3.7: Change "b. Increase in inside diameter" to "b. Increase in inside diameter or decrease in length"

PAGE 40

* 4.5.3.2: At the end, add "Any test fixture used by the contractor must be approved by the government technical agency."

* 4.5.4.2: Delete this paragraph in its entirety

4.5.5: Delete last sentence and substitute: "When ballistically testing the projectile metal parts, reference/calibration components shall be used as a minimum in the target-impact-dispersion phase."

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4.5.5.2.4: Delete this paragraph in its entirety and substitute:

"4.5.5.2.4 Weapon zeroing - boresight and superelevation. Actual superelevation of the weapon shall be measured and recorded for all target dispersion test phases. This requires that the weapon be zeroed at the beginning, end, and during any extended breaks (30 minutes or longer) in target impact test phases using a conventional collimated M26 Pye Watson boresight (or equivalent). Record the weapon elevation and azimuth which aligns the boresight zero on the target center. Then record weapon elevation and azimuth data for all test and calibration phases. The gun superelevation is the weapon elevation during the test phases minus the weapon elevation which produced a muzzle boresight on target center. This data shall be obtained when using either a fixed mount or a service tank weapon system. This information shall be recorded for information only and included in the ballistic test report."

4.5.5.2.4.1: Add new paragraph 4.5.5.2.4.1:

"4.5.5.2.4.1 Gun tube backlash removal. After loading and locking a round in the chamber, the gun tube shall be traversed off the target center, dropped down below the target center, traversed to the opposite side of the target center, elevated to the target center elevation, and traversed onto the target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above or beyond the target center, do not reverse direction of gun tube movement, but repeat the sequence."

PAGE 43

* 4.5.6: At the end, add "Any test fixture used by the contractor must be approved by the government technical agency."

* 4.5.7: At the end, add "Any test fixture used by the contractor must be approved by the government technical agency."

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5.1.1.2: Delete last sentence and substitute: "Pack sixteen (16) fin and boom assemblies arranged in a 4 x 4 pattern, using interlocking separators."

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5.1.1.3: Delete this paragraph in its entirety.

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* 6.11: Delete this paragraph in its entirety and substitute:

"6.11 Probable error factor. The factors shown in Table IV are taken from Table A-21 AMC Pamphlet 706-114 using the 95 percent upper confidence limit ($A_{.05}$) and (N_i-1) degrees of freedon. To define (N_i-1) see 6.5"

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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